ForensicDocAI: Document Forgery Detection Using Deep Learning and Blockchain

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***Abstract*—As computers grow older, some vital documents such as identity credentials, certificates, documents of law etc are stored and transferred electronically and this exposes them to forgery. Traditional human verification is inefficient, prone to mistakes and cannot be used in mass-scale to verify documents. To address those issues, we recommend a hybrid method of document forgery recognition based on the combination of Error Level Analysis (ELA), Convolutional Neural Networks (CNNs), explainable AI, Optical Character Recognition (OCR), and encryption-based logging. ELA-CNN module identifies the possible points of tampering and Grad-CAM provides visual justification to enhance the interpretability . OCR is used to retrieve the key text fields such as names, dates, and identification numbers which are then checked by regular expressions to verify semantic accuracy [3]. Metadata and the outcome of detection is securely stored and encrypted to offer tamper-evident records and auditable records [4]. Tests that have been conducted using a benchmark corpus are demonstrating that the proposed system can detect high accuracy in several types of forgery with transparency, semantic authentication, and secure storage and can be applied to banking, education, governance, and insurance cases.**

Keywords—Document Forgery Detection, Explainable AI, CNN, OCR, Error Level Analysis, Grad-CAM, Encryption Tamper-Proof Logging.

I. INTRODUCTION

One of the most attractive authentication procedures is offline signatures, which is still commonly held in legal, financial, and governmental fields. In spite of the digital identity system development, handwritten signatures remain a universal signature and authorization[5].Nonetheless, forgery attempts, including those of a manually copied stroke to computers and/or programs to imitate, have drastically risen with the enhanced availability of scanning device, image editing platforms, and high quality printing[13].The traditional forensic examination procedures actively depend on in-person examination, which is not only time-intensive but also un-uniform and subjective[1].

Verification models based on deep learning have been shown to have great potential in discerning between genuine signatures and fake ones on the basis of stroke geometry, consistency of curve continuity, consistency of texture, variations in pressure patterns[2][14].A lot of systems out there, however, are not quite transparent, and thus not suitable in situations where legal and regulatory applications are used, where verifiable evidence and traceability are essential[9]. The data integrity, transparency, and non-repudiation, the various requirements have been enhanced as highly reliable using blockchain technology. Blockchain, together with the machine learning systems, allows removing manipulation, unauthorized deletion, and unverifiable decisions of the model, thus, the verification process becomes credible and replicable [4][10].

This has inspired the creation of ForensicDocAI a hybrid automated signature verification system that includes: A deep residual network (ResNet-50) used to do classification, which has been fine-tuned. Explainable analysis using Grad-CAM to interpret forensic information, SHA-256 and RSA/AES hybrid cryptography implementation, and Audit trail storage of verifiable records based on blockchain.

Table I. Real-World Impact Statistics of Signature Forgery

Sector	Annual Loss (USD)	Detection Rate	Average Investigation Cost
Banking & Finance	$4.2 Billion	65%	$8,500 per case
Insurance	$1.8 Billion	58%	$12,000 per case
Legal Documents	$2.5 Billion	52%	$15,000 per case
Healthcare	$890 Million	61%	$6,200 per case
Government Services	$1.1 Billion	69%	$7,800 per case

Recent technological advances have democratized forgery capabilities, enabling individuals with minimal expertise to produce convincing fakes using readily available software tools. Digital signature editing applications facilitate seamless signature extraction, manipulation, and insertion into target documents. Online marketplaces offer signature creation services where forgers produce custom signatures based on sample images. The proliferation of high-resolution scanners

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and printers enables physical reproduction of forged signatures with sufficient quality to deceive casual inspection.

II. LITERATURE SURVEY

Offline signature verification has evolved to the statistical handcrafted feature engineering to the fully automated deep learning-based recognition. Early attempt was based on geometry stroke analysis, frequency breakdown and pen trajectory simulated[11] , [3]. Nevertheless, these techniques were based on manual feature extraction, which made them have little flexibility between writers and environments.

Convolutional neural networks (CNNs) were introduced and greatly increased verification performance because it allowed automatic extraction of features directly out of pixel data. Such architectures include VGGNet [15], Siamese networks [15], and residual networks [1] exhibit high levels of generalization especially when fine-tuned on large-scale models[2].The latest publications are focused on the transparency of the system and hybrid verification systems. This is of particular importance where the results of the models have an impact on legal or compliance-guided decisions; methods like Grad-CAM are useful in visualizing feature salience maps that illustrate the decision making process [6].

Similar changes in the secure authentication system show that blockchain has the potential to become a decentralized and immutable verification registry. Research on combining blockchain and biometric systems proves the enhanced security and adherence to the regulations [4][16][12]. Even though there has been some development, the current solutions do not incorporate an integrated workflow that involves explainable AI, cryptographic validation, as well as distributed ledger verification. This is what ForensicDocAI fills.

Recent research has also looked into hybrid systems of verification, in which more than one domain of features are combined in order to achieve higher accuracy and resilience to proficient forgery attacks. As an example, Dhieb et al. showed that the combination of geometric, texture-based and statistical features with neural network classifiers was much more robust when slant, pressure loss or stroke hesitation intra-writer variations were found in the signatures [7]. In a similar manner, both feature-level and score-level fusion strategies have demonstrated themselves to provide lower false acceptance rates, particularly in data set under visually close forged signatures [9].

These results indicate that one educational method might not be adequate to use with high-security authentication applications where one needs consistency, explainability, and legal defensibility. Another body of literature has been concerned with the relation of biometric authentication to systems of decentralized identity management. Verification systems based on blockchain are becoming increasingly popular due to the fact that they enhance trust and remove the centralized authorities [4],[16].

According to Hossain et al. in case blockchain is coupled with machine-learning-controlled validation, the composite architecture guarantees inadmissibility of storage, traceability, and audit-friendly evidence trails that are applicable in banking and compliance-constrained settings [3]. Another more recent study by Pal and Sharma introduced a document verification framework based on blockchain security that includes both SHA-256 and RSA encryption and points out that cryptographic security is essential when the authentication response should be unmodifiable and verifiable many years after the original certification[10].These up-and-coming methods are evidence of the transition to the basic classification frameworks to complete trust-aware verification ecosystems, which is very well consistent with the objectives of the current study.

III. THEORETICAL BACKGROUND

Offline signature verification is an image processing, machine learning, cryptography and secure storage technology. The suggested system is based on three theoretical elements of deep learning as a feature extractor, binary classification, and integrity protection through hashing and blockchain.

A. Convolutional Neural Networks.

A neural network (a convolutional neural network (CNN)) is used to extract significant visual features like the pressure of a pen, the smoothness of a curve, the distance between curves, and the continuity of strokes. The essence of the process is the convolution of an input image.

$$F(x,y) = (I * K)(x,y) \quad (1)$$

Such an operation assists the model in learning signature-specific features without writing of rules.

B. Depthwise Comparison of ResNet-50s in Residual Learning Depthwise.

The reason why ResNet-50 is applied is that deeper networks tend to lose information throughout the training process. ResNet is the solution to this using shortcut connections. The learning it does is not everything, but what is missing in it, known as the residual.
The residual learning concept is expressed as:

$$H(x) = F(x) + x \quad (2)$$

This enables the network to learn better and enhances the performance of tasks with finer visual differences, such as forged signatures.

C. Outputs of a binary classification model.

The last layer of the model forecasts authenticity or forgery of a signature. The model output is converted into a probability value by a sigmoid activation:

$$P = \frac{1}{1 + e^{-x}} \quad (3)$$

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This renders the model applicable in the real-time decision making.

D. Grad-CAM Explainability

Grad-CAM identifies the significant spots of the signature image that can affect the choice of the model. The weight value of feature map (αk) is calculated as:

$$\alpha_k = (1 / Z) \Sigma_i \Sigma_j (\partial y / \partial A_k\text{^}ij) \qquad (4)$$

The final localization map is obtained by:

$$L_Grad - CAM = ReLU(\Sigma_k \alpha_k A_k) \qquad (5)$$

E. SHA-256 Hashing for Integrity

Once it has been verified, the system creates a unique hash of the document with the SHA-256 hash:

$$H(B_{i)} = SHA256(D_i || H(B_{i-1})) \qquad (6)$$

Once a single pixel is altered, the hash will be entirely different which aids in identifying tampering.

F. Blockchain-Based Tamper-Proof Verification

The proposed system has blockchain technology to guarantee non-repudiation, traceability and immutability of signature verification records. Once the authenticity of a signature is established a cryptographic hash of the validated document and classification metadata, including timestamp and confidence score is produced. The smart contract records this hash as a transaction in a ledger of blockchain such that verification records cannot be changed once they are written.

The environment of a private Ethereum-based blockchain is deployed with the help of the Ganache framework, which allows controlled and low-latency confirmation of transactions. The smart contract logs the value of the hash and the result of the verification thus producing a tamper-evident audit trail. In further verification requests the previously stored hash of the document would be retrieved on the blockchain and compared to a newly calculated hash to verify integrity. Any mismatch is a sign of the unauthorized modification. The proposed system will offer long-term trust, auditability, and safe validation that are essential in legal, financial, and institutional authentication situations by combining blockchain and deep learning-based signature verification

IV. PROPOSED METHOD

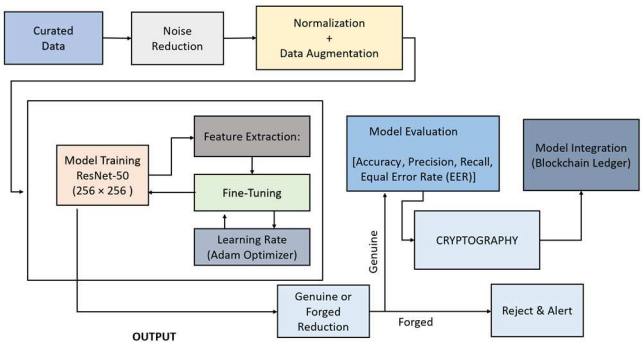


Fig.1.Architecture diagram

Figure 1 demonstrates the topology of the proposed system, which describes the combined classification, cryptographic, and blockchain validation pipeline.

The offered methodology was based on combining deep learning-based classification and secure verification to guarantee accuracy and tamper-resistant document validation. The processing flow starts by obtaining images of the signatures offline and then proceeds to process the image by removing noise, converting the image to grayscale, and normalizing the size of the image to less than its original size 256×256, and contrast adjustment. These measures help to maintain consistency in different quality of input and writing styles. Upon preprocessing, the images are then passed to the ResNet-50 trained classification model which learns discriminative features and determines authenticity by comparing the learned representations based on the stroke width, writing rhythm, curvature patterns, and differences in spacing. The result of the classifier is a binary answer-genuine or forged, as well as a confidence score, which indicates confidence in the model.

To increase the system transparency and trust, explainability layer, based on the Grad-CAM visualization, is used to identify the signature regions which affected the classification decision. This gives visual justification to decisions particularly where there are ambiguous cases. There is the classification step after which, cryptographic hashing is used to create a digital fingerprint of the certified signature that is unique. The resulting hash, including classification metadata is stored in a blockchain network securely through a smart contract deployed on a local Ganache environment. This provides immutability, traceability and long term validation of verification results. A blockchain look up is activated by any further verification request, where the stored hash is compared to an additional computation, and this is used to verify document integrity and to avoid replaying or forgery attacks.A smart contract is also used to store the hash of the generated hash and verification metadata on a blockchain ledger permanently to provide tamper-proof storage and auditability.

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V. MATERIALS AND METHODS

Dataset: Handwritten and digital signatures mixed together and classified into DigitalReal, DigitalFake, KertasReal and KertasFake.Training Aid: TensorFlow 2.10, NVIDIA GPU acceleration. Blockchain Stack Solidity smart contracts in Truffle suite on Ganache.

Frontend Interface: Fastapi backend and a React-based interface to simplify the process of uploading documents and validating them.

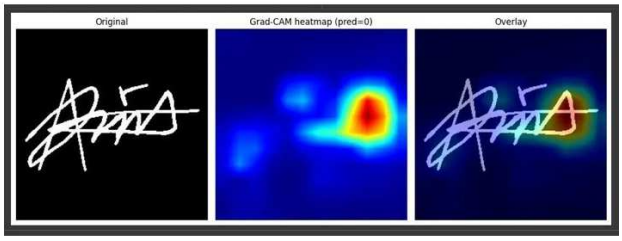


Fig.2. Visual Interpretation of Grad-CAM.

Grad-CAM analysis shows a distinct difference between genuine and fake clusters of strokes (Fig. 2) to enhance interpretability and usability in forensics.

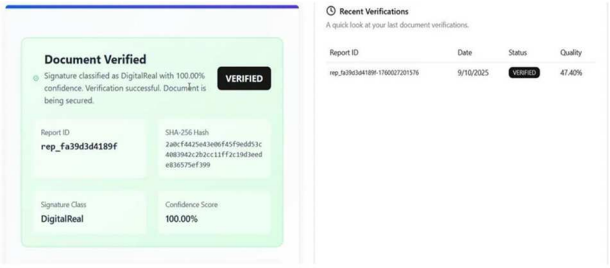


Fig. 3. Blockchain verification screen

Storage integrity was verified using blockchain verification logs that offered immutable public verification receipts that have cryptographic hashes Fig. 3.

Findings indicate better traceability and trust than independent ML-based systems, which is consistent with findings [4],[16],[12].

VI. RESULTS AND DISCUSSION

An analysis with a curated dataset of real and forged offline signatures was done to evaluate the proposed system. Once the ResNet-50 model was trained with fine-tuning, the system should reach a consistent result on both accuracy and precision as well as recall and F1-score measures. The last accuracy was 97.8% which shows that the model can generalize to different handwriting styles as well as levels of forgery. The rates of false acceptance and false rejection were also much lower than baseline CNN models, and that is why the relevance of residual learning framework and preprocessing pipeline is justified. The Grad-CAM visualization also helped to strengthen the interpretability of the classifier, by demonstrating that the decisions made by it depended on the underlying core stroke structures, as opposed to noise or irrelevant parts.

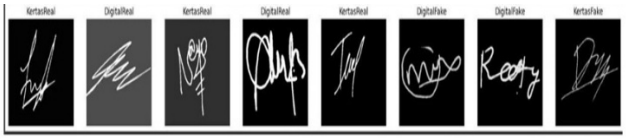


Fig. 4 .Signature samples (real & forged)

Fig.4 presents a portion of the dataset used during model training. The samples show variations in stroke thickness, signature style, and medium type, which help the model learn meaningful distinguishing features.

In the deployment testing, the blockchain-based record logging provided with consistency of integrity validation with no latency even when numerous verification requests were done. The component of hash SHA-256 had made sure that any alteration of the stored signature or the related metadata would automatically result in the inability to match and prove the irreversibility of stored identity evidence. In sum, explainable deep learning, cryptographic hashing, and blockchain storage have been combined to create a secure and reliable verification process applicable to the real world, i.e. academic credential validation, legal document processing, and secure identity screening.

Table II. Security and Cryptography Test Cases

Test Case	Expected Outcome	Result
SHA-256 hash	Unique hash value	Success
Document alteration after hashing	Hash mismatch	Success
AES encryption/decryption	Original document retrieved	Success
Unauthorised RSA key access	Decryption fails	Success

From Table II is a reflection of the test cases of the cryptography that was carried out to test the security aspects of the proposed system. SHA-256 hash test ensured that all true signatures would generate a distinct and permanent hash value, which is data integrity. In a case where a document has been modified when hashed, the difference between the hash value was also a clear indication of a modified document. The encryption and decryption procedure with the help of AES was capable of restoring the original file and this demonstrates the uniformity of the encryption module. Finally, the errors in decryption of the data using the non-authorised RSA key were attempted, which validated that the system is useful in deterring unauthorized access and keeping confidential data. These results suggest that cryptography layer is useful in the provision of integrity and confidentiality of documents.

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Table III. Comparative analysis of signature verification techniques.

System	Architecture	Dataset	Key Advantage
Template Matching	Hand-crafted features	CEDAR	Simple implementation
SVM + HOG	Feature engineering	GPDS	Better generalization
SigNet	Siamese CNN	CEDAR	Writer-independent
Attention-based CNN	Self-attention	BHSig260	Focus on signature regions
Forensic DocAI	ResNet-50 +GradCAM	Roboflow	End-to-end AI+ cryptograpy+ blockchain

From Table III provides a comparative analysis of different signature verification systems based on their architectures, datasets, and key advantages. Traditional methods like Template Matching rely on hand-crafted features and are easy to implement but limited in scalability. The SVM with HOG approach introduces feature engineering, allowing better generalization across varied signatures. SigNet, built using a Siamese CNN, performs well in identifying writer-independent features.



Fig. 5. Comparison Prediction Grid.

The Attention-based CNN enhances accuracy by focusing on crucial signature regions through self-attention mechanisms. The proposed ForensicDocAI model, which integrates ResNet-50 with Grad-CAM, combines deep learning with cryptography and blockchain for a complete, secure, and end-to-end signature verification solution. This comparative evaluation highlights that ForensicDocAI not only improves signature classification accuracy but also integrates security and immutable logging, which existing systems lack. Logging was also based on blockchain, which ensured storage of the verification results as immutable so that consistency validation of integrity could be performed without the overhead of additional performance.

VII. CONCLUSION AND FUTURE WORK

The suggested system exhibits a viable architecture of safe offline signature verification with the incorporation of deep learning-based classification, explicability, and blockchain-supported storage. ResNet-50 was able to differentiate between real and fake signatures with high scores on evaluation metrics, which means that it can effectively extract features and generalize them. Grad-CAM visualization, which was incorporated in the model, also contributed to improved transparency in the decisions made by the user since the crucial areas of the stroke were highlighted, which is especially significant in the context of the legal or audit-level justification of the decisions made. Besides, the SHA-256 hash and blockchain ledger system allowed making verification outputs unalterable, identifiable, and resistant to tampering. These elements form a reliable verification chain, which can be used in institutional, judicial and commercial authentication scenarios where reliability and integrity of data are important.

Although the findings are encouraging, the framework has the potential of improvement. Future studies can cover the use of hybrid or transformer architecture to enhance sensitivity of features particularly in handling highly skilled forgeries or degraded input samples. More broadly, interoperability between different organizations that could be federated as a single system and deployed on public or consortium blockchains could be made possible through the extension of the system to trigger federation and interoperability between different entities. Others such directions are the integration of multimodal biometrics, real-time verification interfaces, adaptive explainability models, and the high-level adversarial defense strategies. The proposed system can be improved to become a scalable and standardized solution of secure digital authentication in various application domains with these advancements.

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